Hanzo 4.0 — GPU-Native AI Compute Network with DEX+EVM+FHE Triumvirate

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Hamiltonian LLMs at L1 GPU-Residency from chain to model

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Abstract

On 14 February 2026 the Hanzo network activated its 4.0 epoch. The network became a sovereign Layer 1 running the canonical Lux-family triumvirate (DEX + EVM + FHE) on a GPU-native Quasar 4.0 stack, with Hamiltonian LLMs as a first-class L1 service and Hanzo Market Maker (HMM) as the native DEX for AI compute. This paper documents the launch as a retrospective: the four-version chronology (1.0 initial AI compute network, 2.0 Hamiltonian LLMs and the AI Coin, 3.0 full post-quantum, 4.0 GPU-native), the AI Chain Virtual Machine (AIVM) opcode surface, the HMM cross-listing with Lux D-Chain and Zoo DEX through the Liquidity Protocol common API, the GPU-Residency invariant from chain to model, and the on-mainnet receipts captured at activation block height $h_0 = 36{,}963{,}000$.

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1 Chronology

Hanzo, founded by the Techstars '17 batch, is now four major versions deep. Each successor strictly subsumes its predecessor; nothing in the 1.0 line was deprecated.

Version	Active from	Theme	Defining changes
1.0	2018–2023	Initial AI compute network	Centralised inference and training infrastructure with off-chain billing.
2.0	2024	Hamiltonian LLMs + AI Coin	On-chain accounting for inference, the AI Coin native asset, the first version of the HMM DEX.
3.0	2025	Full PQ	ML-DSA mandatory for validators; Hanzo settles under Lux Quasar 3.0 (LP-105) cert lanes; FHE inference research preview.
4.0	2026-02-14	GPU-native + native DEX with Liquidity Protocol	Sovereign L1 with the canonical 9-chain template (LP-134), AIVM GPU-resident, HMM cross-listed via Liquidity Protocol, FHE inference promoted to production (`hanzo-fhe-inference`).

Table 1: Hanzo major-version timeline. The 4.0 boundary co-activates with the Lux 4.0 boundary so that the family hard-fork is atomic.

2 The Triumvirate on Hanzo

The *triumvirate* is the three-pillar set of DEX + EVM + FHE that defines a complete financial-rail-resident L1. Hanzo 4.0 is the canonical *triumvirate network*: its chain set *is* exactly those three subsystems — DEX (HMM), EVM (Cancun-equivalent C-Chain template), and FHE (confidential AI inference) — and nothing else. Hanzo's EVM is plain EVM; AIVM is the dedicated A-Chain VM at the Lux primary network (LP-134, attestation + AI provenance), which Hanzo accesses *natively* (§2.1), not as a Hanzo-local clone.

The other Lux-family networks carry the financial-rail core too, but as a subset of broader chain sets: Lux 4.0 runs the full 9-chain topology (LP-134); Zoo 4.0 adds per-LLM chains, conservation chains, and L1→L4 tenancy; Pars 2.0 adds P/X/S-Chain plus Persian-diaspora precompiles. Hanzo is the focused triumvirate where every chain is the financial rail; everything non-financial is serviced by Lux primary natively.

Hanzo 4.0 also serves as the **Rust canonical node** of the family: `~/work/hanzo/cli` (Rust) and the `hanzo-libs` workspace under `~/work/hanzo/node` are the reference Rust implementations the other networks integrate against.

2.1 Native Lux primary-chain access

Hanzo, like every Lux-family L1, natively uses the Lux primary network's chains — P, C, X, Q, Z, A, B, M, F — without bridging. "Native" means same Quasar cert lanes, same subject-binding, same finality, no wrapped assets, no cross-chain hop tax. Hanzo specifically reaches into Lux primary for:

- **A-Chain (AIVM)** for attestation, model-registry anchors, and AI-provenance receipts. The 8 AIVM precompiles (`0x0a01`–`0x0a08`) live on *Lux* A-Chain; Hanzo's HMM marketplace records AI-compute attestations there directly.
- **Z-Chain (ZVM)** for ZK rollups when Hanzo customers need proof-aggregated batches.
- **M-Chain (MVM)** for any MPC ceremony Hanzo's HMM requires (CGGMP21, FROST, Ringtail-general).
- **B-Chain (BVM)** for canonical cross-ecosystem bridging.
- **F-Chain (FVM)** when Hanzo's local FHE service needs to delegate to Lux primary's FHE substrate.

Hanzo does not duplicate any of those chains locally. The boundary between Hanzo's three chains and Lux's primary nine is cert-lane subject-binding, not a bridge.

2.2 DEX — Hanzo Market Maker (HMM)

HMM is the native AI-compute exchange, originally introduced in HIP-8 (2024) and graduated to L1 status at the 4.0 boundary. It quotes, matches, and settles five compute primitives:

1. **Inference seconds.** GPU-second futures by model class (LLM-7B, LLM-70B, LLM-405B, ViT-B, etc.), priced per quality tier.
2. **Training slots.** Fixed-window training capacity on specific GPU classes, with deterministic data-locality routing.
3. **GPU leases.** Bare-metal capacity by hour, expressed through the AIVM `GPU_LEASE` opcode (`0x0a05`) on Lux A-Chain, accessed natively from Hanzo's EVM.
4. **Embedding throughput.** Tokens/sec for retrieval-augmented generation pipelines.
5. **Active-inference budgets.** Hamiltonian LLM compute budgets (cf. HIP-7).

The price function for inference seconds in pool p at time t is the constant-product invariant adapted for capacity decay:

$$x_t \cdot y_t = k_p \cdot \exp(-\lambda_p \cdot (t - t_0)) \,, \tag{1}$$

where x_t is unsold compute capacity, y_t is the AI-coin reserve, k_p is the genesis liquidity, and λ_p is the capacity-decay rate calibrated per pool to express that compute time not used by t is forever lost. Equation 1 is solved on-GPU within the EVM execution path with native AIVM hooks; round-trip latency for a swap is under one millisecond.

Liquidity Protocol cross-listing. HMM registered with the Liquidity Protocol common settlement API (LP-310) at $h_0 + 12$. The router exposes HMM pools to Lux D-Chain and Zoo DEX, allowing a single transaction to draw inference capacity quoted on Hanzo against collateral held on Lux or Zoo.

2.3 EVM — C-Chain template, GPU-native

Hanzo's EVM is plain Cancun-equivalent EVM (the C-Chain template from LP-134), GPU-native via cevm v0.41+. AIVM is the dedicated A-Chain VM at the Lux primary network and is *not* a Hanzo local; Hanzo uses Lux A-Chain (AIVM) natively for AI-provenance opcodes. The eight AIVM precompiles (`0x0a01`–`0x0a08`) live on Lux A-Chain. From Hanzo's EVM, calls into AIVM look like any other native primary-chain access (no bridge, same cert lanes, same subject-binding):

Address	**Function**	**Notes**
`0x0a01`	HMM_LOOKUP	Resolve model_root → metadata
`0x0a02`	INFER_REQUEST	Submit inference job to market
`0x0a03`	INFER_VERIFY	Verify TEE quote + receipt
`0x0a04`	TRAIN_BOUNTY	Post training bounty
`0x0a05`	GPU_LEASE	Lease GPU pool capacity
`0x0a06`	AGENT_AUTH	Authorize agent action under capability
`0x0a07`	F_CALL	Cross-chain call to F-Chain (FHE)
`0x0a08`	M_CUSTODY	Cross-chain call to M-Chain (MPC)

Table 2: AIVM precompile surface (Lux A-Chain, accessed natively by Hanzo). All GPU-resident at the 4.0 boundary.

GPU-residency is enforced by the same QuasarSTM 4.0 transactional memory boundaries used on Lux:

Invariant 2.1 (GPU residency, LP-137)**.** *For every transaction τ executed at height $h \geq h_0$, the canonical execution path runs on a GPU device g with all input, intermediate, and output state resident on g's HBM for the duration of τ. Any host-RAM round-trip during τ is a violation.*

The AIVM precompiles `0x0a02` (INFER_REQUEST) and `0x0a03` (INFER_VERIFY) are direct-call paths into QuasarGPU's inference kernels — the same hardware that performs cert aggregation also serves the model. Co-location halves end-to-end latency for AI-bound contracts.

2.4 FHE — Confidential AI Inference

Confidential AI inference is documented in `hanzo-fhe-inference`. The 4.0 launch promotes that paper from research preview to production. Three application patterns activate at h_0:

1. **Encrypted prompt inference.** Clients submit CKKS-encrypted prompts; Hanzo validators run the model over ciphertexts; only the client (holder of the secret key) can decrypt. Hanzo never sees the plaintext.

2. **Encrypted decision-tree inference.** TFHE-encrypted feature vectors traversed against TFHE-encrypted decision boundaries; useful for medical and financial classification.

3. **Encrypted federated aggregation.** Replaces the secure-aggregator trust assumption in K-party fine-tuning with homomorphic gradient summation.

The FHE substrate is reached through the AIVM `F_CALL` precompile (`0x0a07`). The bridge to Lux F-Chain (LP-013 v2) is the canonical fallback for clients that prefer Lux-rooted threshold decryption.

3 $AI Coin Tokenomics + GPU Mining (PoAI)

The AI Coin (ticker `AI`) is the native asset of the Hanzo L1. Tokenomics are documented in detail in `hanzo-tokenomics`; the 4.0 boundary preserves all parameters and adds two emission streams.

- **HMM fee routing.** 0.30% of HMM swap volume routes to the canonical Lux-family split: 50% liquidity providers, 30% emission stabiliser, 15% ve-AI lockers, 5% DAO treasury.
- **FHE relay rewards.** Capped at 0.5% annualised AI supply, paid to validators that perform FHE bootstrapping and ciphertext maintenance. The cap sits inside the previously-allocated ecosystem bucket; *no new emission.*

Proof-of-AI (PoAI) GPU mining. Hanzo's emission curve is funded by Proof-of-AI: validators must prove they served verifiable inference work (TEE quote + receipt, verified via `INFER_VERIFY`) to earn block rewards above the floor. PoAI is a strict superset of proof-of-stake; validators with no inference receipts still earn the floor.

Theorem 3.1 (Supply preservation)**.** *The 4.0 emission additions preserve the fixed total supply of 10^9 AI coins. The FHE relay reward cap and HMM fee routing changes are funded entirely from previously-allocated buckets in the `hanzo-tokenomics` schedule.*

Proof. Mechanical. The 2.0-era schedule allocates 30% of total supply (3×10^8) to ecosystem incentives. The FHE reward cap is 0.5% of 10^9 for at most 8 years $= 4 \times 10^7$. The HMM liquidity-bootstrap vault is 5×10^7. Both fit within 3×10^8. $\square$

4 HMM and ASO Protocols

HMM is paired with ASO (Active Sequence Orchestration), the inference-routing protocol that selects which validator serves each request. ASO and HMM together form the Hanzo compute marketplace.

4.1 ASO routing

For an inference request r with budget b and quality target q, ASO selects the validator v^* minimising the objective

$$v^* = \arg\min_{v \in V} \ \alpha_1 \cdot \mathrm{lat}_v(r) + \alpha_2 \cdot \mathrm{cost}_v(r) + \alpha_3 \cdot \mathrm{qualitygap}_v(r, q) + \alpha_4 \cdot \mathrm{distance}_v(r), \tag{2}$$

subject to $\mathrm{cost}_{v^*}(r) \le b$. The weights $\alpha_1 \dots \alpha_4$ are governance parameters; default values are fixed at $(0.4, 0.3, 0.2, 0.1)$ at h_0.

4.2 HMM-ASO settlement

A served inference produces a TEE-signed receipt $\rho = (r,\, v^*,\, t_{\mathrm{start}},\, t_{\mathrm{end}},\, \mathrm{cost})$. The HMM contract verifies ρ via the `INFER_VERIFY` precompile and atomically:

1. Debits the buyer's pool position by cost.
2. Credits the validator's settlement account.
3. Updates the pool's capacity decay (§ 2, eq. 1).

4. Emits a Liquidity-Protocol event so cross-listed pools on Lux and Zoo see the price update.

The cycle finishes inside a single block.

5 Bridge to Lux, Zoo, Pars

Counterparty	Mechanism	Proof	Settlement
Lux	A-Chain attestation (`lux-achain-attestation`)	ML-DSA-87	24 s
Zoo	Liquidity Protocol router	ML-DSA-87	30 s
Pars	Liquidity Protocol router	ML-DSA-87	30 s
Ethereum	Optimistic light client	ECDSA + ML-DSA-87	7 d

Table 3: Hanzo 4.0 cross-chain bridges. All bridge proofs are post-quantum at the signature layer.

6 Activation Receipts

Event	Height	Notes
Quasar 4.0 epoch boundary	h_0	cert lanes 1–4
QuasarSTM 4.0 activation	h_0	STM root rotated
GPU-Residency invariant on	$h_0 + 1$	violations = 0
HMM genesis pools	$h_0 + 6$	12 inference, 5 training, 8 GPU-lease
Liquidity Protocol register	$h_0 + 12$	router id `lp.hanzo.router`
First $AI \to LUX$ HMM swap	$h_0 + 47$	cross-listed via Lux D-Chain
First $AI \to ZOO$ HMM swap	$h_0 + 81$	cross-listed via Zoo DEX
First confidential inference	$h_0 + 122$	CKKS-encrypted prompt
F-Chain bridge live	$h_0 + 220$	via M-Chain custody
First active-inference job	$h_0 + 350$	Hamiltonian LLM, HIP-7 path

Table 4: Selected mainnet receipts from Hanzo 4.0 activation.

6.1 Hardware footprint at h_0

Indicator	Value at $h_0 + 24h$
Active validators	312
Aggregate GPU FLOPs (FP16)	$4.2 \cdot 10^{18}$
Inference TPS sustained	2,100,000
Median inference latency (LLM-7B)	42 ms
Median HMM swap latency	0.91 ms

Table 5: Production hardware footprint observed in the 24h after h_0.

7 Conclusion

Hanzo 4.0 closes the loop on what 1.0 began in 2018: a verifiable, liquid market for AI compute, settled on chain, served on validator GPUs, priced by an AMM that respects capacity decay. The triumvirate makes the market complete — HMM provides the price, AIVM provides the logic, the

FHE substrate provides confidentiality. Liquidity Protocol exposes HMM pools to the rest of the family, and the same hardware that votes in Quasar 4.0 cert lanes serves the model.

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